



# **Artificial Intelligence**

## **Intelligent Reading Comprehension and Quiz Generation System using Machine Learning & Neural Networks**

### **Final Project Report**

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# Abstract

This project is a Reading Comprehension (RC) Quiz System built using traditional machine learning techniques and the RACE dataset. The system can automatically generate quiz questions from reading passages, create multiple-choice options, check user answers, give hints, and show performance analytics through an interactive web application.

The project contains two main models. Model A is responsible for question generation and answer verification. It generates questions from important sentences in the passage using template-based methods. Logistic Regression, Support Vector Machine (SVM), and a hard-voting ensemble are used to verify whether an answer option is correct or incorrect. K-Means clustering and Label Propagation are also implemented to include unsupervised and semi-supervised learning techniques. A Random Forest ranker is used to select the best generated question.

Model B is responsible for distractor and hint generation. It creates incorrect answer choices (distractors) and generates hints to help users solve the quiz step by step.

The system is developed using Python, Scikit-learn, Pandas, and Streamlit. Different evaluation metrics such as Accuracy, Precision, Recall, Macro F1-score, confusion matrices, purity score, and silhouette score are used to evaluate the models. The final system provides an interactive and educational quiz platform using classical machine learning methods.

## Introduction & Motivation

### Introduction

Reading comprehension systems are widely used in educational platforms to help students improve their understanding, reasoning, and learning skills. Multiple-choice reading comprehension quizzes are especially useful because they test a student's ability to understand passages, identify important information, and choose correct answers from different options. Creating such quizzes manually, however, requires significant time and effort from teachers.

This project presents a Reading Comprehension (RC) Quiz System developed using traditional machine learning techniques and the RACE dataset. The system automatically generates quiz questions from reading passages, creates distractor options, verifies user answers, provides hints, and displays evaluation analytics through an interactive web application. The project focuses on using classical machine learning approaches.

The system consists of two major components. Model A handles question generation and answer verification using Logistic Regression, Support Vector Machine (SVM), hard-voting ensemble learning, K-Means clustering, and Label Propagation. Model B handles distractor and hint generation to improve the quiz experience and make the questions more interactive.

## Motivation

The main motivation behind this project is to reduce the manual effort required to create educational quizzes and to explore how traditional machine learning techniques can be applied to reading comprehension tasks. Many modern systems rely heavily on large neural network models, which require high computational resources and long training times. In contrast, this project demonstrates that classical machine learning methods can still be used to build useful and interactive educational systems.

## Related Work

Automatic question generation and reading comprehension systems have been widely studied in the field of Natural Language Processing (NLP). Researchers have proposed different approaches for generating questions, distractors, and educational quizzes from textual passages.

1- Zhang et al. presented a comprehensive literature review on Machine Reading Comprehension, discussing major datasets, evaluation techniques, and machine learning approaches used in reading comprehension systems.

Reference: [Application of machine reading comprehension techniques for named entity recognition in materials science - PMC](#)

2- Du et al. proposed a neural question generation system for reading comprehension using sequence-to-sequence learning models. Their work showed that automatically generated questions can be grammatically correct and contextually relevant for educational tasks.

Reference: [\[1705.00106\] Learning to Ask: Neural Question Generation for Reading Comprehension](#)

3- Gao et al. introduced a difficulty-controllable question generation framework that generates reading comprehension questions with different difficulty levels while maintaining question quality and relevance.

Reference: [\[1807.03586\] Difficulty Controllable Generation of Reading Comprehension Questions](#)

4- Ghanem et al. proposed the HTA-WTA framework for inferential reading comprehension question generation. Their system focused on generating deeper comprehension questions

instead of only simple extractive questions from passages.

Reference: [Question Generation for Reading Comprehension Assessment by Modeling How and What to Ask - ACL Anthology](#)

5- Kumar et al. proposed a machine learning based distractor generation system for multiple-choice reading comprehension questions. Their work focused on generating realistic distractor options that improve quiz quality and student engagement.

Reference: [On the application of Transformers for estimating the difficulty of Multiple-Choice Questions from text - ACL Anthology](#)

## Dataset Analysis

The RACE (Reading Comprehension Dataset From Examinations) dataset was used in this project for training and evaluation. RACE is a large-scale reading comprehension dataset collected from English examination passages designed for middle school and high school students. The dataset contains passages, questions, multiple-choice options, and correct answers.

Each sample in the dataset includes:

- A reading passage (article)
- A comprehension question
- Four answer options (A, B, C, D)
- The correct answer label

The dataset was selected because it is widely used for evaluating reading comprehension systems and contains diverse question types that test reasoning and understanding abilities.

## Dataset Structure

The original dataset contains the following important columns:

- Articles
- Questions
- Options (A,B,C,D)
- Answers

## Data Preprocessing

Several preprocessing steps were applied before training the models:

- Converted all text to lowercase

- Removed punctuation marks
- Combined article, question, and answer options into a single text feature for Model A verification
- Created train, dev , and test splits

The dataset was divided into:

- 80% Training set
- 10% Development set
- 10% Test set

```
train = pd.read_csv('data/processed/train.csv')
dev   = pd.read_csv('data/processed/dev.csv')
test  = pd.read_csv('data/processed/test.csv')
print ('Train: ',train.shape)
print('Dev: ',dev.shape)
print('Test: ',test.shape)]
```

[45] Python

```
... Train: (61506, 9)
    Dev: (13180, 9)
    Test: (13180, 9)
```

train.head(3)

|   | id             | article                                           | question                                 | A                            | B                                            | C                                  | D                                       | answer | combined text                                     |
|---|----------------|---------------------------------------------------|------------------------------------------|------------------------------|----------------------------------------------|------------------------------------|-----------------------------------------|--------|---------------------------------------------------|
| 0 | middle4234.txt | Ann is my classmate. She comes from New York, ... | What is Ann's hope ?                     | To get fat.                  | To have more friends.                        | To speak Chinese well.             | To live in New York.                    | C      | ann is my classmate she comes from new york an... |
| 1 | high326.txt    | The bicycling craze came in when we were just     | How did the writer feel about tricycles? | They were very hard to ride. | Riding tricycles was an enjoyable amusement. | They were not as fast as bicycles. | They were not proper for women to ride. | A      | the bicycling craze came in when we were just ... |

## Feature Preparation

For Model A answer verification, option-level training was used. Each option was converted into a separate training sample using the format:

article + question + option

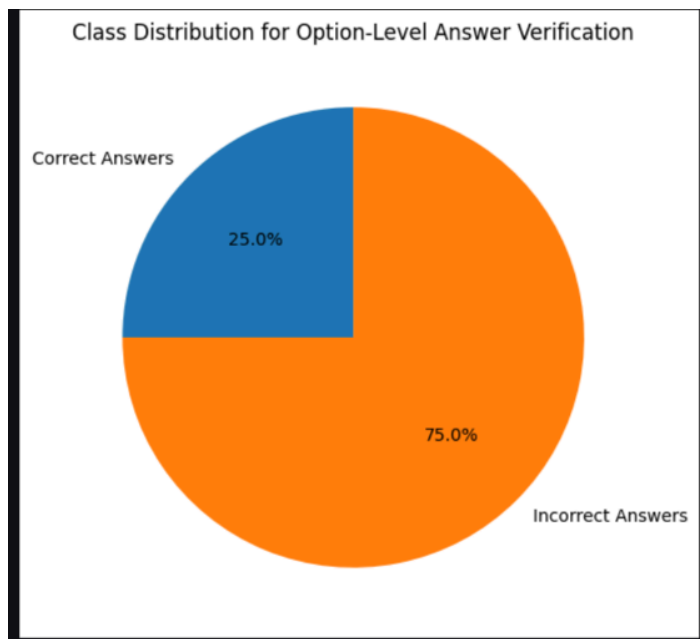
The correct option was labeled as:

- 1 - Correct answer
- 0 - Incorrect answer

This created an imbalanced dataset because each question contains:

- 1 correct option
- 3 incorrect options

To handle this imbalance, class weighting techniques were applied during model training.



## Text Representation

The text data was first converted into numerical form using CountVectorizer. A binary One-Hot Encoding approach was used, where the model checks whether certain words or word pairs are present in the text.

The vectorizer was configured to:

- use the 10,000 most important words,
- include both single words (unigrams) and two-word combinations (bigrams),
- remove common English stop words such as “the”, “is”, and “and”,
- ignore words that appear too frequently in almost all documents.

This process helps convert textual passages, questions, and options into numerical features that can be used by traditional machine learning models such as Logistic Regression and SVM.

## **Dataset Challenges**

Some challenges observed in the dataset include:

- Class imbalance due to option-level training
- Long reading passages
- Semantic similarity between correct answers and distractors
- Difficulty in capturing deep comprehension using traditional ML methods

Despite these challenges, the dataset provides a suitable benchmark for evaluating classical machine learning approaches for reading comprehension quiz systems.

## **Model A: Design, Training, Results**

Model A is responsible for two main tasks: generating a question from a reading passage and verifying whether the selected answer option is correct or incorrect. The model follows a traditional machine learning approach.

For answer verification, the system uses option-level training. This means that each answer option is treated as a separate sample. The article, question, and one option are combined together and given to the model. The model then predicts whether that option is correct or incorrect.

### **Design**

For each question, the dataset contains four answer options: A, B, C, and D. Since only one option is correct, the correct option is labeled as 1, while the remaining three incorrect options are labeled as 0.

The input format used for training was:

article + question + option

The following traditional machine learning models were implemented:

- Logistic Regression
- Support Vector Machine (SVM)
- Hard Voting Ensemble

- K-Means Clustering
- Label Propagation
- Random Forest Question Ranker

Logistic Regression and SVM were used for supervised answer verification. The Hard Voting Ensemble combined the predictions of Logistic Regression and SVM. K-Means was used as an unsupervised learning method, while Label Propagation was used as a semi-supervised learning method.

## Question Generation

For question generation, a template-based approach was used. First, the system selects an important sentence from the passage by comparing it with the correct answer. Then, question templates are applied to generate possible questions.

For example, if the selected sentence contains the correct answer, the answer is replaced with a blank and converted into a question. A Random Forest classifier is then used as a question ranker to select the most suitable generated question.

## Training Process

The dataset was first preprocessed and converted into numerical features using CountVectorizer. Binary One-Hot-Encoding representation was used with unigram and bigram features. The models were trained on the training set and evaluated on the development set. Class weighting was used during training because the dataset is imbalanced. In option-level training, each question has one correct option and three incorrect options, meaning that incorrect answers appear more frequently than correct answers.

## Results

|                      | Accuracy (%) | Macro F1 | Precision | Recall |
|----------------------|--------------|----------|-----------|--------|
| Logistic Regression  | 51.6         | 0.4801   | 0.2599    | 0.5065 |
| SVM                  | 51.6         | 0.4802   | 0.26      | 0.5069 |
| Hard Voting Ensemble | 51.63        | 0.4803   | 0.2599    | 0.5062 |

- Logistic Regression achieved an accuracy of 51.60% with a Macro F1-score of 0.4801. The model showed balanced recall for both classes but struggled with precision because of class imbalance.



- SVM produced results similar to Logistic Regression because both models used the same One-Hot Encoding feature representation. The model achieved 51.60% accuracy and a Macro F1-score of 0.4802.
- The hard-voting ensemble combined Logistic Regression and SVM predictions and achieved the best performance with 51.63% accuracy.

| MODEL A — UNSUPERVISED / SEMI-SUPERVISED |                |                     |                     |
|------------------------------------------|----------------|---------------------|---------------------|
| K-MEANS SILHOUETTE                       | K-MEANS PURITY | LABEL PROP ACCURACY | LABEL PROP MACRO F1 |
| <b>0.0095</b>                            | <b>0.7913</b>  | <b>0.7793</b>       | <b>0.4552</b>       |

- K-Means clustering achieved a purity score of 0.7913, showing moderate separation between clusters.
- Label Propagation achieved higher accuracy but lower Macro F1-score because of dataset imbalance.

| Ensemble confusion matrix |                     |                   |  |
|---------------------------|---------------------|-------------------|--|
|                           | Predicted Incorrect | Predicted Correct |  |
| Actual Incorrect          | 13698               | 12663             |  |
| Actual Correct            | 4339                | 4448              |  |

The confusion matrix shows that the model was able to predict both correct and incorrect classes, but it still struggled due to the difficulty of reading comprehension and the imbalance between correct and incorrect options.

## Model B: Design, Training, Results

Model B is responsible for distractor generation and hint generation in the reading comprehension quiz system. Distractors are incorrect answer options that are added along with the correct answer to create multiple-choice questions. The purpose of Model B is to generate meaningful distractors and provide hints that help users solve questions step by step.

## Design

For each question, the correct answer is first identified from the dataset. Then, distractor candidates are selected from the passage and ranked using machine learning techniques.

The system generates:

- One correct answer
- Three distractor options

The distractors are designed to:

- be related to the passage,
- avoid exact duplication of the correct answer.

The final quiz contains four options:

- 1 correct answer
- 3 distractors

## Hint Generation

Model B also generates hints to help users answer questions gradually. The hints are generated from important sentences in the reading passage.

Three levels of hints are provided:

1. General clue
2. More specific clue
3. Nearly direct clue

Users can reveal hints one by one before revealing the final answer.

## Training Process

The textual data was converted into numerical form using CountVectorizer with One-Hot-Encoding representation. The distractor ranking model was trained using article text and answer options.

The system classifies:

- correct answers,
- distractor options.

Traditional machine learning methods were used to rank and select the best distractors for each question.

## Results

| MODEL B — DISTRACTOR RANKER |              |              |              |
|-----------------------------|--------------|--------------|--------------|
| ACCURACY                    | MACRO F1     | PRECISION    | RECALL       |
| <b>51.2%</b>                | <b>0.476</b> | <b>0.756</b> | <b>0.516</b> |

The model achieved moderate accuracy while generating meaningful distractors for quiz questions. The higher precision value indicates that the model was relatively effective at identifying distractor options correctly.

| Model B confusion matrix |                          |                      |  |
|--------------------------|--------------------------|----------------------|--|
|                          | Predicted Correct Answer | Predicted Distractor |  |
| Actual Correct Answer    | 4390                     | 4397                 |  |
| Actual Distractor        | 12768                    | 13593                |  |

The confusion matrix shows that the model was better at identifying distractor options than correct answers. This is expected because the dataset contains more distractors than correct answers.

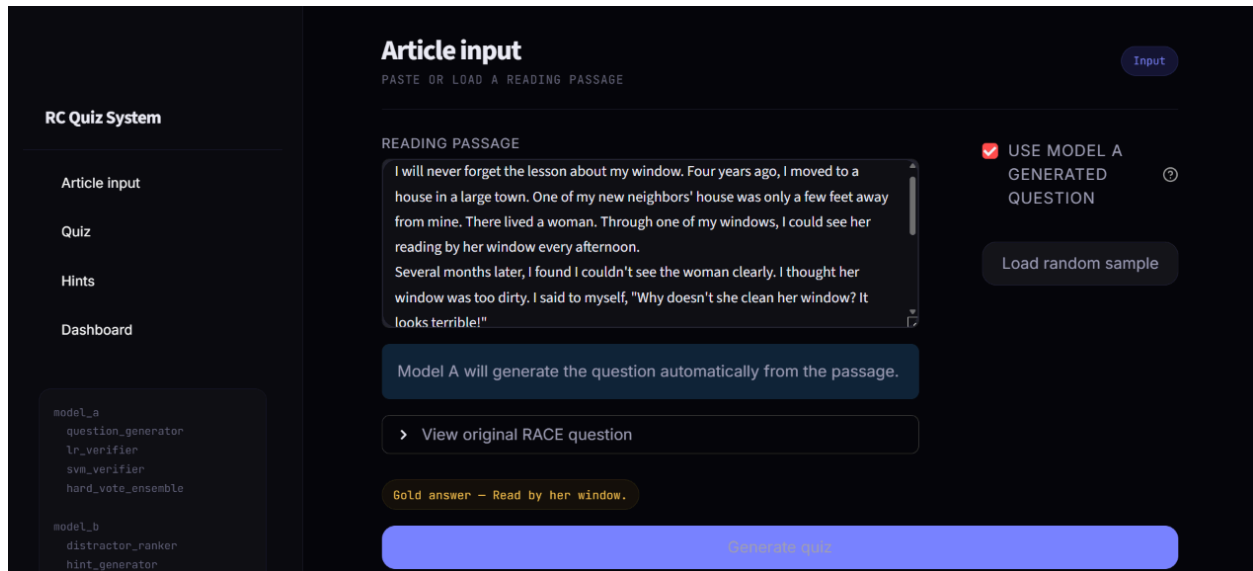
## User Interface Description

The Reading Comprehension Quiz System provides an interactive web-based interface developed using Streamlit. The interface allows users to generate quizzes from reading passages, answer multiple-choice questions, receive hints, and view model evaluation analytics.

The user interface is divided into four main sections:

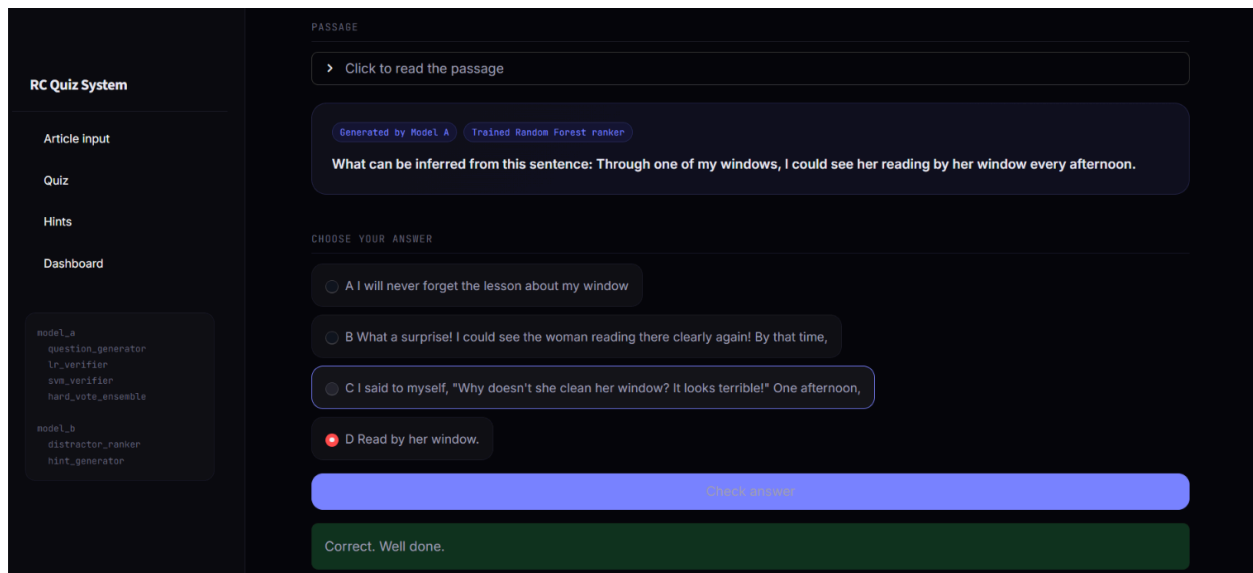
### Article Input Page

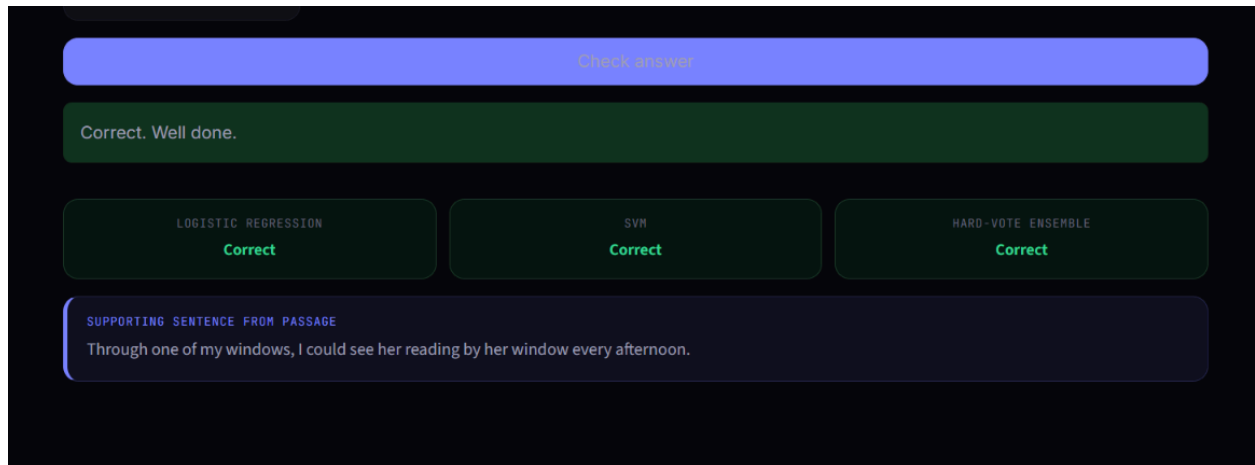
The Article Input page allows users to either paste their own reading passage or load a random sample from the RACE dataset. Users can also provide a custom question and correct answer, or allow Model A to automatically generate a question from the passage. After entering the required information, the user can click the “Generate Quiz” button to create the quiz.



## Quiz Page

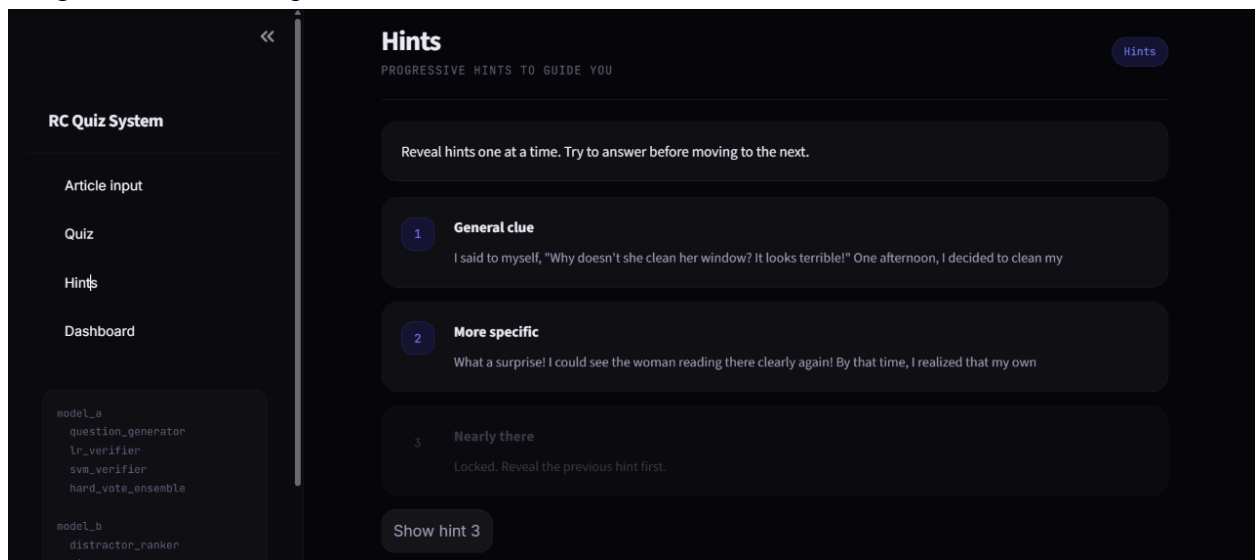
The Quiz page displays the reading passage, generated question, and four multiple-choice options. Users can select an answer and click the “Check Answer” button to verify their response. The system then displays whether the answer is correct or incorrect along with predictions from Logistic Regression, SVM, and the hard-voting ensemble model. An explanation sentence from the passage is also shown to support the answer.

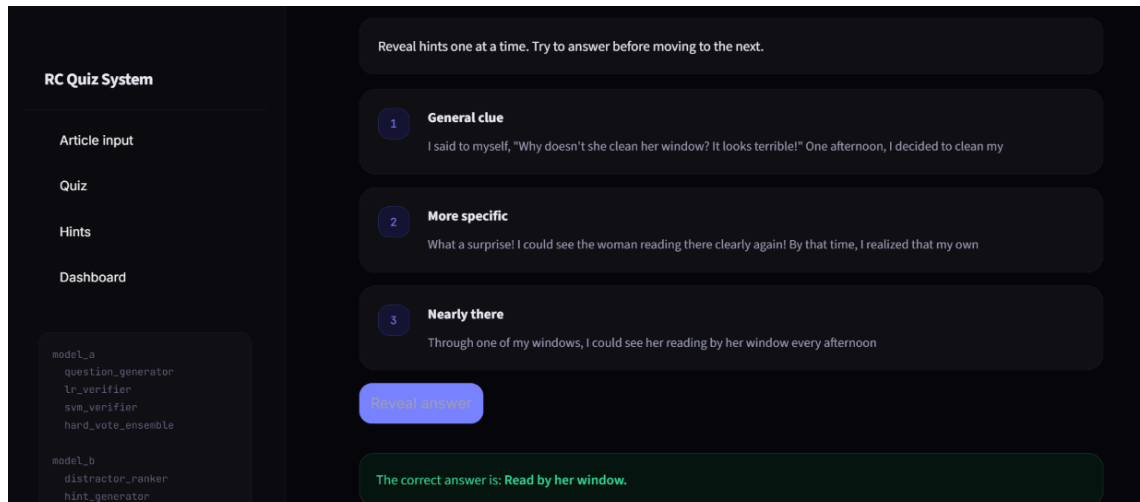




## Hints Page

The Hints page provides step-by-step hints generated by Model B. Users can reveal hints gradually to help solve the question before revealing the final answer. The hint system is designed to make the quiz more interactive and educational.





## Analytics Dashboard

The Analytics Dashboard displays model evaluation results and live session statistics. It provides detailed information about:

- supervised learning results,
- unsupervised learning results,
- semi-supervised learning results,
- confusion matrices,
- and user session analytics.

The dashboard includes:

- Accuracy
- Precision
- Recall
- Macro F1-score
- K-Means purity and silhouette score
- Label Propagation metrics
- Session logs and CSV export

**Main features:**

- Model comparison tables
- Confusion matrices
- Live session metrics
- Exportable session logs

# Dashboard

MODEL METRICS AND SESSION ANALYTICS

Stats

## LIVE SESSION

TOTAL ATTEMPTS

1

EXACT MATCHES

1

USER ACCURACY

100.0%

AVG. VERIFY LATENCY

0.005s

## MODEL A - VERIFIER METRICS

ENSEMBLE ACCURACY

51.63%

MACRO F1

0.480

PRECISION

0.260

RECALL

0.506

## Model comparison

|                      | Accuracy (%) | Macro F1 | Precision | Recall |
|----------------------|--------------|----------|-----------|--------|
| Logistic Regression  | 51.6         | 0.4801   | 0.2599    | 0.5065 |
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## Ensemble confusion matrix

|                  | Predicted Incorrect | Predicted Correct |
|------------------|---------------------|-------------------|
| Actual Incorrect | 13698               | 12663             |
| Actual Correct   | 4339                | 4448              |

## MODEL A - UNSUPERVISED / SEMI-SUPERVISED

K-MEANS SILHOUETTE

0.0095

K-MEANS PURITY

0.7913

LABEL PROP. ACCURACY

0.7793

LABEL PROP. MACRO F1

0.4552

## MODEL B - DISTRACTOR RANKER

ACCURACY

51.2%

MACRO F1

0.476

PRECISION

0.756

RECALL

0.516

## Model B confusion matrix

|                       | Predicted Correct Answer | Predicted Distractor |
|-----------------------|--------------------------|----------------------|
| Actual Correct Answer | 4390                     | 4397                 |
| Actual Distractor     | 12768                    | 13593                |

## SESSION LOG

|   | question                                                                            | question_source      | selected            | correct             | gold_label |
|---|-------------------------------------------------------------------------------------|----------------------|---------------------|---------------------|------------|
| 0 | What can be inferred from this sentence: Through one of my windows, I could see her | Generated by Model A | Read by her window. | Read by her window. | 1          |

Export session log to CSV

## Evaluation & Discussion

The Reading Comprehension Quiz System was evaluated using multiple machine learning metrics because it combines supervised, unsupervised, and semi-supervised learning techniques.

For Model A answer verification, Accuracy, Precision, Recall, Macro F1-score, and confusion matrices were used. The Hard Voting Ensemble achieved the best results with an accuracy of 51.63% and a Macro F1-score of 0.4803. Due to class imbalance in the option-level dataset, Macro F1-score and confusion matrices were important for evaluation. The models could identify correct and incorrect answers but struggled with deep semantic understanding of long passages. Logistic Regression and SVM produced similar results because both used One-Hot Encoding features.

In unsupervised learning, K-Means clustering achieved a purity score of 0.7913, while the semi-supervised Label Propagation model achieved higher accuracy but lower Macro F1-score because of dataset imbalance.

Model B was evaluated using Accuracy, Precision, Recall, and Macro F1-score for distractor classification. The system successfully generated meaningful distractors, hints, and quiz options. The Streamlit interface improved usability by providing interactive quizzes, explanations, and analytics dashboards.

Overall, the project demonstrated that traditional machine learning techniques can effectively build a functional and educational reading comprehension quiz system, despite limitations in deep semantic understanding compared to neural networks.

## Limitations & Future Work

Although the Reading Comprehension Quiz System works well for quiz generation and answer checking, it still has some limitations. The machine learning models gave only average accuracy because traditional methods like Logistic Regression and SVM cannot fully understand the meaning of long passages. They mostly learn from word patterns instead of understanding context deeply.

Another problem is dataset imbalance because each question has one correct answer and three wrong answers. This makes the models learn more from incorrect options. The question generation system also uses templates, so some questions may look repetitive or less natural. Similarly, the distractors (wrong options) may sometimes be too easy or unrealistic.



The system also depends a lot on preprocessing and feature extraction because deep learning models were not used. In the future, the project can be improved by using better feature techniques like TF-IDF, word embeddings, and contextual representations. Better balancing methods can also improve model performance.

The quiz and distractor generation modules can be improved with more diverse templates and smarter ranking methods. Extra features like difficulty levels, progress tracking, adaptive quizzes, and user accounts can also make the system more useful for learning. Future versions may compare traditional machine learning methods with modern transformer models like BERT and T5 to achieve better results.

## **Conclusion**

This project developed a Reading Comprehension Quiz System using traditional machine learning techniques and the RACE dataset. The system can automatically generate quizzes, check user answers, create distractor options, provide hints, and show performance analytics through a Streamlit interface.

Model A handled question generation and answer verification using methods like Logistic Regression, SVM, ensemble learning, K-Means clustering, and Label Propagation. Model B focused on generating distractors and hints to make quizzes more interactive and useful for users.

The project showed that traditional machine learning methods can still be used effectively for reading comprehension tasks without deep learning models. Even though the models achieved moderate accuracy, the system successfully completed quiz generation and answer verification tasks.

The Streamlit interface made the system interactive and user-friendly by allowing users to practice quizzes, view hints, and analyze model performance through dashboards and evaluation metrics.

Overall, the project successfully combined supervised, unsupervised, semi-supervised, and ensemble learning techniques into a complete educational quiz platform for automated reading comprehension practice.

## References

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